

Tester for 555 Timer and 741 Op-amp ICs

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Sometimes you do not get proper output from a circuit due to faulty ICs. This circuit can test timer 555 and op-amp 741 ICs, which are commonly used in projects, to save you from this problem.

Circuit and working

Fig. 1 shows circuit diagram of the tester which is built around the very 555 timer (IC1) and the op-amp 741 (IC2) that are to be tested. IC1 is wired as astable multivibrator with output

frequency of around 1Hz. Resistors R1 and R2 and capacitor C1 are the timing components that decide output frequency. You can change 1Hz frequency by changing values of these components.

The 555 tester section is basically an LED flasher circuit. Square-wave output at pin 3 of IC1 drives LED1 and LED2 which glow alternately. That is, when pin 3 is low, LED1 glows and when pin 3 is high, LED2 glows. This indicates that the 555 timer under test is in good condition. If the 555 is faulty, both the LEDs may



remain off or glow dimly, or one or both of them may glow continuously.

PARTS LIST

Semiconductors:

- IC1 - 555 timer
- IC2 - 741 op-amp
- LED1-LED3 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 68k-ohm
- R2 - 39k-ohm
- R3, R4, R7 - 1.2k-ohm
- R5, R6 - 4.7k-ohm

Capacitors:

- C1 - 10 μ F, 25V electrolytic
- C2 - 10nF ceramic disk

Miscellaneous:

- BATT.1 - 9V battery
- CON1 - 2-pin connector
- S1, S2 - On/off switch
- ZIF1, ZIF2 - 8-pin ZIF socket
- 2-pin terminal connector

Test Points

Test point	Details
TP0	0V
Across TP0 and TP1	9V when switch S1 is closed
Across TP0 and TP2	Around 1Hz
Across TP0 and TP3	Around 8V when switch S2 is closed

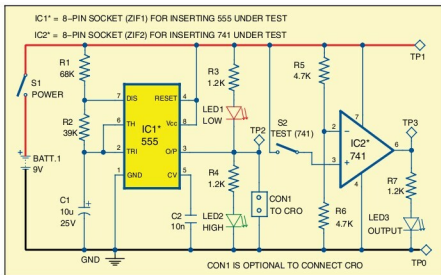


Fig. 1: Circuit diagram of 555 and 741 ICs tester

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